Robust Estimation and Filtering in the Presence of Bounded Noise

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Abstract—**In this note optimal algorithms for robust estimation and filtering are constructed. The noise is considered a deterministic variable belonging to a set described by a norm. Previous results, obtained for complete (one-to-one) and approximate information [1], are now extended to partial and approximate information. This information seems useful in dealing with dynamic systems not completely identifiable and/or with two different sources of noise, for example, process and measurement noise. For different norms characterizing the noise, optimal algorithms (in a min-max sense) are shown. In particular, for Hilbert norms a linear optimal algorithm is the well-known minimum variance estimator. For l_∞ norm an optimal algorithm, computable by linear programming, is presented.**

State estimation is formalized in the context of the general theory. For stable systems, an approximate state estimation is obtained by neglecting higher order powers. An upper bound estimating the approximation introduced is derived. A numerical example illustrates the application of the theory.

I. Introduction

In recent years an alternative approach to the classical statistical one has been developed for system identification. Generally, in estimation theory the uncertainty is considered a random variable described by a (partially) known density function. In the worst case approach no statistical assumption is available or used and the noise is considered a deterministic variable belonging to a certain set [1]–[4]. A relationship between optimal estimators for statistical and worst case can be found in [5].

In this note we extend previous results [1] obtained for complete (one-to-one) and approximate information to the case of partial (not necessarily one-to-one) and approximate information. The use of partial and contaminated information seems useful for two problems in system identification: first, for problems in which two different sources of noise should be considered, for example, process and observation noise of a dynamic system; secondly, for problems in which the parameters are not completely identifiable from the output measurements and some *a priori* information is given.

We construct optimal algorithms defined in the sense of information-based complexity (see [1]–[4]). Now we briefly discuss the subject of this theory. We are interested in approximating $Sf \in Z$ where S is a given linear mapping $S:F \to Z$, and f belongs to a certain subset $K \subset F$ representing the *a priori* information available about it (f and S are called, respectively, problem element and solution operator). The element f is not exactly known but only approximate information $h = Nf + \rho$, $h \in H$ is given, where N is called information operator and the noise ρ is unknown but bounded by a given constant ϵ. An approximation to Sf is provided by an operator φ (called algorithm or estimator) operating on the information h. Optimal algorithms minimize the maximal distance between the actual solution Sf and the computed solution $\varphi(h)$ for the worst problem element f and for the worst information h. The error of an optimal algorithm is called intrinsic error or radius of information. Strongly optimal algorithms are optimal algorithms for any fixed information h. These formal definitions are exposed in Section II.

In Section III we show optimal algorithms for some norms describing the uncertainty of the problem elements f and of the noise ρ. First, we consider the uncertainty bounded by Hilbert norms. This problem has been studied in [6]. The linear optimal algorithm presented has the same structure of the maximum *a posteriori* estimator for Gaussian distributions, but depends on a "smoothing factor" whose practical computation is not an easy task. In this section we show that Gauss–Markov estimator is an optimal algorithm if the bound ϵ on the noise ρ is sufficiently small (corresponding to the statistical case when the knowledge on the covariance matrix of the noise is essential). If ϵ is large (equivalent to a poor knowledge of the covariance matrix) an optimal algorithm is 0. Then, we turn our attention to other norms in the measurement space. For l_∞ norm a nonlinear strongly optimal algorithm, computable by linear programming is derived in [1]. Here, with the same norm, we show a linear optimal algorithm using the optimal information (see [7]) minimizing the intrinsic error.

In Section IV we consider the problem of state estimation of a dynamic system in the presence of unknown but bounded noise. This problem is addressed in [8]–[11]. In particular, in [10] a recursive algorithm, having the same structure of Kalman filter, is shown. Unfortunately, the solution suggested is not sharp if instantaneous constraints are considered. In the same paper an approximate solution is presented deriving equations of an ellipsoidal bound of the set of possible states. For dynamic systems with additive and parametric uncertainty belonging to polyhedral sets, Sankaran and Barmish [11] show that the convex hull of the true set can be propagated forward in time.

In this note state estimation is formalized in the context of the general theory previously presented. The algorithms given in Section III provide optimal solutions for different norms characterizing the uncertainties. For stable systems, in order to reduce the computational complexity, an approximate solution is obtained by neglecting higher order powers. An upper bound of the approximation error is derived. A numerical example, involving a second-order dynamic system, concludes the note.

II. Definitions and Notation

Let F be a linear normed n-dimensional space over the real field and let K be the unit ball in F defined by

$$K = \{f \in F : \|f\| \le 1\}. \tag{1}$$

Consider a given linear operator S, called a solution operator, which maps F into Z

$$S : F \to Z$$

where Z is a linear normed r-dimensional space over the real field. Our aim is to estimate an element Sf of the Z space knowing approximate and partial information about the element $f \in K$. Define a linear operator N, called information operator, which maps F into a linear normed m-dimensional space H

$$N : F \to H.$$

In this note we do not assume that N is a complete information, i.e., N is not necessarily a one-to-one mapping. This means that the problem element f may be not identifiable from the knowledge of Nf. In general, in the presence of noise, exact information Nf about f is not available and only perturbed information h is given. In this context we assume that the uncertainty of information $\rho = h - Nf$ is unknown but bounded

$$\|h - Nf\| \le \epsilon \tag{2}$$

where ϵ is some fixed number. An algorithm φ is a mapping (in general nonlinear) from H into Z

$$\varphi : H \to Z,$$

i.e., it provides an approximation $\varphi(h)$ of Sf using the available information. Define the following set

$$T(h) = \{f \in K : \|h - Nf\| \le \epsilon\}. \tag{3}$$

For simplicity we assume that $T(h)$ is a nonempty set. For each approximate information $h \in H$ we define a local error $E(\varphi, h)$ of an

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algorithm φ as

$$E(\varphi, h) = \sup_{f \in T(h)} \|Sf - \varphi(h)\|, \qquad \forall h \in H. \tag{4}$$

An algorithm φ_s is called strongly optimal if

$$E(\varphi_s, h) = \inf_{\varphi} E(\varphi, h), \qquad \forall h \in H. \tag{5}$$

The strong optimality is a meaningful property in estimation problems as it minimizes the difference between the actual solution Sf and the estimated solution $\varphi(h)$ for the worst element f belonging to the set $T(h)$ for any fixed approximate information h. Note that strongly optimal algorithms map h into the center of a minimal ball containing the set $S\{T(h)\}$, $\forall h \in H$. For this reason they are often called central algorithms (see [1]–[4]).

The (global) error $E(\varphi)$ of an algorithm φ is defined as

$$E(\varphi) = \sup_{h \in H} E(\varphi, h). \tag{6}$$

An algorithm φ_o is called (globally) optimal if

$$E(\varphi_o) = \inf_{\varphi} E(\varphi). \tag{7}$$

The minimum global error $E(\varphi_o)$ is called intrinsic error or radius of information.

III. Optimal Estimators for Linear Problems

A. Measurement Space Equipped with Hilbert Norms

This problem is studied in [6] when F, H, and Z are Hilbert spaces, not necessarily of finite dimensions. Let F and H be equipped with the following Hilbert norms:

$$\|f\|^2 = f^T V f; \qquad \|h\|^2 = h^T R h \tag{8}$$

where V and R are symmetric, positive definite matrices. Then, based on [6], one can derive the optimal algorithm of the form

$$\varphi_o(h) = S\left(N^T R N + \frac{\mu\epsilon^2}{1-\mu} V\right)^{-1} N^T R h \tag{9}$$

where μ is a parameter between zero and one, which is solution of the equation

$$\sup_{f \in T(0)} \|Sf\| = \sup_{\mu\|f\|^2 + (1-\mu)\epsilon^{-2}\|Nf\|^2 \le 1} \|Sf\|. \tag{10}$$

Note that (9) is the maximum *a posteriori* estimation (minimum variance and minimum absolute error) of Sf when f and ρ are independent random variables normally distributed with mean value and covariance matrix given by

$$M_f = M_\rho = 0; \; \Sigma_f = \left(\frac{\mu\epsilon^2}{1-\mu} V\right)^{-1}; \qquad \Sigma_\rho = R^{-1}.$$

Unfortunately, from a practical point of view, the computation of the smoothing coefficient μ, i.e., the solution of the equation (10), is not an easy task. However, for some specific cases, the parameter μ can be easily found. We consider two such cases depending on the size of the bound ϵ of the noise.

Theorem 1: Let F, H, and Z be Hilbert spaces. Let F and H be equipped with the norms (8). Then a linear optimal algorithm is given by

$$\varphi_o(h) = \begin{cases} S(N^T R N)^{-1} N^T R h & \text{if } T(0) \subset K; \\ 0 & \text{if } T(0) \supseteq K. \end{cases} \tag{11}$$

Proof: It is easy to prove that $\mu = 0$ for the first case, $T(0) \subset K$, and $\mu = 1$ for the second one, $T(0) \supseteq K$.

Let $T(0) \subset K$. Note that this condition implies that N is one-to-one. Under this assumption we obtain

$$\sup_{f \in T(0)} \|Sf\| = \sup_{\|Nf\| \le \epsilon} \|Sf\|. \tag{12}$$

From this we conclude that $\mu = 0$ is a solution of (10) and Gauss–Markov is an optimal algorithm.

If $T(0) \supseteq K$ we get

$$\sup_{f \in T(0)} \|Sf\| = \sup_{\|f\| \le 1} \|Sf\|. \tag{13}$$

Thus, $\mu = 1$ is now a solution of (10). Then, for μ tending to one, the algorithm φ_o in (9) tends to zero. Hence, the zero algorithm is optimal as claimed. △

The interest of Theorem 1 lies in the fact that, if $T(0) \subset K$, a linear optimal algorithm is Gauss–Markov estimator. Note that if N is one-to-one and $K \equiv F$ Gauss–Markov is strongly optimal (see [12]). If the knowledge on the noise is weak, i.e., $T(0) \supseteq K$ (corresponding to the statistical case when a poor knowledge of the covariance matrix of the noise is available), the optimal algorithm is 0.

B. Measurement Space Equipped with l_∞ Norms

In this section we present a linear optimal algorithm when H and Z are equipped with l_∞^w weighted norms[1] and K is a piecewise linear, convex, and balanced set. The use of the l_∞^w weighted norm in the measurement space allows one to handle situations in which the bound on the noise may be different at each time. In the following, we assume that S is a linear functional. Under this assumption the intrinsic error is given by (see [4])

$$E(\varphi_o) = \sup_{f \in T(0)} |Sf|. \tag{14}$$

Define an $(n + m)$ by n matrix P and an $(n + m)$-dimensional vector d as

$$P = [I, N]^T; \; d = [0, h]^T \tag{15}$$

where I is the identity matrix and 0 is the null vector of appropriate dimensions. If K is the unit ball, $T(h)$ can be rewritten as

$$T(h) = \{f \in F : \max_{1 \le i \le (m+n)} w_i |P_i f - d_i| \le \epsilon\} \tag{16}$$

where $w_1 = \cdots = w_n = \epsilon$ and $w_{n+1}, \cdots, w_{n+m}$ are the weights defined in the H space. Consequently, the set $T(0)$ results

$$T(0) = \left\{f \in F : |P_i f| \le \frac{\epsilon}{w_i}; \; i = 1, \cdots, (m+n)\right\} \tag{17}$$

and the intrinsic error is

$$E(\varphi_o) = \sup_{\substack{f \in F: |P_i f| \le \epsilon/w_i \\ i = 1, \cdots, (m+n)}} Sf = \sup_{\substack{f \in F: |P_i f| = \epsilon/w_i \\ i = t_1, \cdots, t_n}} Sf. \tag{18}$$

The constraints $t_1, \cdots, t_n$ are called active constraints of the linear programming problem (18). Define the n by n matrix $\tilde{P}$ containing the rows $t_1, \cdots, t_n$ of the matrix P and the n-dimensional vector $\tilde{d}$ containing the measurements at times $t_1, \cdots, t_n$ of the vector d

$$\tilde{P} = [P_{t_1}, \cdots, P_{t_n}]^T; \; \tilde{d} = [d_{t_1}, \cdots, d_{t_n}]^T. \tag{19}$$

The information $\tilde{d}$ is the optimal information in the sense defined in [7] and [13]. Note that at most n measurements out of m should be available and are used by the following optimal algorithm.

Theorem 2: Let F and H be linear normed spaces equipped with l_∞^w norms and S a linear functional. If rank $\tilde{P} = n$ the algorithm

$$\varphi_o(h) = S\tilde{P}^{-1}\tilde{d} \tag{20}$$

is linear and optimal.

[1] The norm $\|h\|_\infty^w$ is defined as $\|h\|_\infty^w = \max_i w_i |h_i|$; $w_i > 0$.

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Proof: Using (16), the local error of the algorithm (20) is

$$E(S\tilde{P}^{-1}\tilde{d}, h) = \sup_{f:\|Pf-d\|_\infty^w \le \epsilon} |Sf - S\tilde{P}^{-1}\tilde{d}|. \tag{21}$$

Since

$$\{f : \|Pf-d\|_\infty^w \le \epsilon\} \subseteq \{f : \|\tilde{P}f-\tilde{d}\|_\infty^w \le \epsilon\} \tag{22}$$

it results that

$$E(S\tilde{P}^{-1}\tilde{d}, h) \le \sup_{f:\|\tilde{P}f-\tilde{d}\|_\infty^w \le \epsilon} |Sf - S\tilde{P}^{-1}\tilde{d}|$$

$$= \sup_{f:\|\tilde{P}(f-\tilde{P}^{-1}\tilde{d})\|_\infty^w \le \epsilon} |S(f-\tilde{P}^{-1}\tilde{d})| = \sup_{f:\|\tilde{P}f\|_\infty^w \le \epsilon} |Sf|. \tag{23}$$

Using (18) and (19) we obtain

$$E(S\tilde{P}^{-1}\tilde{d}, h) \le E(\varphi_o). \tag{24}$$

Since the opposite inequality is obvious, the proof is complete. △

The condition about the invertibility of the matrix $\tilde{P}$ is not restrictive in practice; in fact, if rank $\tilde{P} < n$ the intrinsic error is not finite. In [1] the optimality properties of an algorithm similar to (20) are studied for complete and approximate information.

Remark: The previous theorem can be easily extended when S is a linear operator and Z is equipped with l_∞ norm. In this case the active constraints of the problem can be computed componentwise and a linear optimal algorithm is the rectangular matrix r by m having row by row the optimal solution of each component of Sf obtained by Theorem 2. △

Remark: Central nonlinear algorithms can be easily constructed when F, H, and Z are equipped with polyhedral norms. Unfortunately, for norms different than l_∞ the number of constraints is exponential, and therefore the computational complexity is high. However, these algorithms can be used in practical problems where the number of measurements is limited. △

IV. State Estimation of Linear Systems in the Presence of Bounded Noise

In this section we apply the general theory previously developed to state estimation of linear systems when process and observation noise are unknown but bounded.

Consider the following discrete, linear, time-invariant dynamic system represented in state variable form

$$\begin{cases} x(i+1) = Ax(i) + Bu(i) \\ y(i) = Cx(i) + \eta(i) \end{cases} \qquad i = 0, 1, \cdots, k \tag{25}$$

where $x(i)$ is the r-dimensional state vector; $y(i)$ is the q-dimensional observation vector; $u(i)$ and $\eta(i)$ are, respectively, the p-dimensional process noise and the q-dimensional observation noise vectors; A, B, and C are given matrices with compatible dimensions.

At time k the problem element f is defined as

$$f = [x(0), u(0), \cdots, u(k-1)]^T. \tag{26}$$

It results that the space F of the elements f is a linear space of dimensions $n = r + pk$. The subset K contains the information about the initial conditions (if available) and the process noise. Since our goal is to estimate the state $x(k)$, solution operator $Sf = x(k)$ is given by

$$S = (A^k \; A^{k-1}B \; A^{k-2}B \; \cdots \; AB \; B) \tag{27}$$

and information operator N results as

$$N = \begin{pmatrix} C & 0 & 0 & \cdots & 0 & 0 \\ CA & CB & 0 & \cdots & 0 & 0 \\ CA^2 & CAB & CB & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ CA^{k-1} & CA^{k-2}B & CA^{k-3}B & \cdots & CB & 0 \\ CA^k & CA^{k-1}B & CA^{k-2}B & \cdots & CAB & CB \end{pmatrix}. \tag{28}$$

The space Z of the solution Sf is linear and r-dimensional. The exact information Nf is corrupted by observation noise and only the approximate information h is available

$$h = [y(0), \cdots, y(k)]^T = Nf + [\eta(0), \cdots, \eta(k)]^T. \tag{29}$$

The space H is linear and of dimensions $m = q(k+1)$. According to (2), we assume that the uncertainty of information

$$\rho = [\eta(0), \cdots, \eta(k)]^T \tag{30}$$

is unknown but bounded by a constant ϵ. An optimal estimate of the state $x(k)$ is provided by the algorithm (11) or by the algorithm (20)

$$\hat{x}(k) = \varphi_o(h). \tag{31}$$

The estimate $\hat{x}(k+1)$ is simply given by $\hat{x}(k+1) = A\hat{x}(k)$.

Unfortunately, the dimension of the matrices S and N increases when k increases and the use of the algorithms (11) and (20) is practically unfeasible for k large. Hence, for stable systems, in order to reduce the computational complexity, we look for an algorithm approximating an "easier" solution $\tilde{S}\tilde{f}$ obtained by neglecting the higher powers in (27) and (28). The "easier" problem element $\tilde{f}$ is defined as

$$\tilde{f} = [u(k-j), \cdots, u(k-1)]^T; \qquad j < k. \tag{32}$$

Consequently, the approximate solution and information are

$$\tilde{S} = (A^{j-1}B \; A^{j-2}B \; \cdots \; AB \; B) \tag{33}$$

$$\tilde{N} = \begin{pmatrix} CB & 0 & \cdots & 0 & 0 \\ CAB & CB & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ CA^{j-2}B & CA^{j-3}B & \cdots & CB & 0 \\ CA^{j-1}B & CA^{j-2}B & \cdots & CAB & CB \end{pmatrix}. \tag{34}$$

Note that the number of rows in (34) is chosen such that $\tilde{N}$ is a full rank matrix. The perturbed information $\tilde{h} = \tilde{N}\tilde{f} + \tilde{\rho}$ is

$$\tilde{h} = [y(k-j+1), \cdots, y(k)]^T. \tag{35}$$

However, j in formula (32) should be appropriately chosen in order that $\tilde{S}\tilde{f}$ is "sufficiently" close to Sf. The next theorem shows how to compute j for stable systems in order to guarantee a given approximation γ. By $\sigma(A)$ we denote the spectral radius of the matrix A.

Theorem 3: Let $\sigma(A) < 1$. If $\|A^j\| \le \gamma < 1$ then

$$\lim_{k\to\infty} \|Sf - \tilde{S}\tilde{f}\| \le \|B\| \frac{\gamma}{1-\gamma} \sum_{i=0}^{j-1} \|A^i\|. \tag{36}$$

Proof: Note that

$$\|Sf - \tilde{S}\tilde{f}\| = \left\| A^k x(0) + \sum_{i=j}^{k-1} A^i Bu(k-i-1) \right\|$$

$$\le \|A^k x(0)\| + \sum_{i=j}^{k-1} \|A^i Bu(k-i-1)\|. \tag{37}$$

Since $x(0)$ and $u(i)$ are in K, we obtain

$$\|Sf - \tilde{S}\tilde{f}\| \le \|A^k\| + \|B\| \sum_{i=j}^{k-1} \|A^i\| \tag{38}$$

and

$$\lim_{k\to\infty} \|Sf - \tilde{S}\tilde{f}\| \le \|B\| \sum_{i=j}^{\infty} \|A^i\| \le \|B\| (\|A^j\| + \|A\| \cdot \|A^j\| + \cdots$$

$$+ \|A^{j-1}\| \cdot \|A^j\| + \|A^j\| \cdot \|A^j\| + \|A\| \cdot \|A^j\| \cdot \|A^j\| + \cdots). \tag{39}$$

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Since $\|A^j\| \le \gamma$, we get

$$\lim_{k\to\infty} \|Sf-\tilde{S}\tilde{f}\| \le \|B\|(\gamma+\gamma^2+\cdots)(1+\|A\|+\|A^2\|+\cdots+\|A^{j-1}\|)$$
$$= \|B\| \frac{\gamma}{1-\gamma} \sum_{i=0}^{j-1} \|A^i\| \quad (40)$$

and the proof is complete. △

The constant γ should be chosen according to precision and computational requirements. It is to remark that γ can be sufficiently small since in (33) and (34) just the higher order powers are neglected. When j is chosen a complete solution to state estimation can be obtained by using the optimal algorithms (11) and (20).

Example: Consider the dynamic system given by

$$\begin{cases} x(i+1) = \begin{pmatrix} 0 & 1 \\ -.01 & -0.2 \end{pmatrix} x(i) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u(i) \\ y(i) = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} x(i) + \eta(i) \end{cases}.$$

Assume that F, H, and Z are Hilbert spaces and $R = I$. Since condition $T(0) \subset K$ is satisfied, the Gauss-Markov estimator is an optimal algorithm. Considering the following data

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
$y_1(k)$	0.30	−0.50	0.10	0.00	−0.10	0.20
$y_2(k)$	0.10	−0.90	0.20	−0.10	0.00	0.40

the optimal estimates computed by using relation (11) are

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$\hat{x}_1(k)$	−0.50	0.10	0.00	−0.10	0.20	0.00
$\hat{x}_2(k)$	0.09	−0.02	0.00	0.02	−0.04	0.00

Choosing $\gamma = 10^{-4}$ relation $\|A^j\|_2 \le \gamma$ is satisfied for $j = 6$. The optimal approximate estimate for $i \ge 4$ is

$$\hat{x}(i+1) = \begin{pmatrix} 0.00 & 0.08 & 0.07 & -0.08 & -0.23 & 0.15 & 0.61 & -0.38 & -1.62 & 1.00 \\ 0.00 & -0.02 & -0.01 & 0.01 & 0.04 & -0.03 & -0.12 & 0.07 & 0.31 & -0.20 \end{pmatrix} \begin{pmatrix} y(i-4) \\ y(i-3) \\ y(i-2) \\ y(i-1) \\ y(i) \end{pmatrix}.$$

Using the available data we obtain

	$k=5$	$k=6$
$\hat{x}_1(k)$	0.25	0.08
$\hat{x}_2(k)$	−0.05	0.02

An upper bound on the asymptotic difference between true and approximate solutions can be easily computed by means of Theorem 3

$$\lim_{k\to\infty} \|Sf-\tilde{S}\tilde{f}\| \le 2.3\times 10^{-4}.$$

V. Conclusions

In this note problems of robust estimation and filtering of dynamic systems in the presence of unknown but bounded measurement and process noise are studied. These uncertainties are not only described by ellipsoidal-type constraints but also an instantaneous bound on each measurement is considered. In both cases linear optimal algorithms (in a min-max sense) are shown. An asymptotic approximate estimate of the state of the dynamic system is easily computed and an upper bound on the approximation introduced is derived. A numerical example, showing that this error is neglectable, concludes the note.

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On the Expectation of the Product of Four Matrix-Valued Gaussian Random Variables

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***Abstract*—The formula for the expectation of the product of four scalar real Gaussian random variables is generalized to matrix-valued (real or complex) Gaussian random variables. As an application of the**

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